

An Explainable Federated Learning Framework for Privacy-Preserving Pneumonia Detection Using Chest X-ray Images

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Abstract—Deep learning models for pneumonia detection are commonly trained using centrally pooled medical data, an approach that conflicts with patient privacy regulations and does not reflect the distributed nature of clinical data across healthcare institutions. This paper presents an explainable federated learning framework for privacy-preserving pneumonia detection using a Squeeze-and-Excitation (SE) enhanced DenseNet121 model trained collaboratively across three simulated hospital clients without sharing patient images. Two federated optimization strategies, FedAvg and FedProx, are evaluated under identical training settings, while Grad-CAM is employed to provide visual explanations of model predictions by highlighting clinically relevant lung regions. The proposed framework is trained and internally evaluated on the RSNA Pneumonia Detection Challenge dataset and further assessed, without fine-tuning, on an independent Labeled Chest X-ray Images dataset to examine cross-domain generalization. On the RSNA test set, FedAvg achieves an accuracy of 82.49% with an AUC of 88.52%, whereas FedProx achieves an accuracy of 82.16% with an AUC of 88.63% and improves recall from 73.28% to 77.27%. Under external evaluation, FedAvg attains high precision but its recall decreases to 26.15%, while FedProx maintains a substantially higher recall of 54.62% at a comparable AUC. These findings demonstrate that incorporating the FedProx optimization strategy improves robustness and generalization under heterogeneous data distributions while preserving patient privacy and providing interpretable predictions through Grad-CAM.

Index Terms—Federated learning, explainable artificial intelligence, pneumonia detection, chest X-ray imaging, DenseNet121, squeeze-and-excitation networks, Grad-CAM, medical image analysis.

I. INTRODUCTION

Pneumonia remains a leading cause of hospitalization and mortality worldwide [1], with chest X-ray imaging serving as the primary diagnostic tool due to its low cost and widespread availability. However, interpreting chest radiographs is time-consuming and prone to inter-observer variability. Recent advances in deep learning have significantly improved automated pneumonia detection, with DenseNet121 emerging as a strong

baseline following CheXNet [2], which builds on the densely connected architecture proposed in [3].

Most existing approaches rely on centralized training, requiring medical images from multiple institutions to be collected in a single repository. Such data sharing is often restricted by patient privacy regulations and institutional policies. Federated Learning (FL), introduced by McMahan *et al.* [4], enables collaborative model training without exchanging patient data. While FedAvg is the standard aggregation algorithm, its performance may degrade under heterogeneous data distributions. FedProx [5] addresses this limitation by incorporating a proximal regularization term that improves training stability.

Despite preserving data privacy, federated models remain difficult to interpret. Explainable AI techniques such as Grad-CAM [6] improve model transparency by highlighting image regions that influence predictions, enabling clinicians to verify whether the model focuses on clinically relevant lung regions.

To address these challenges, this paper proposes an explainable federated learning framework for pneumonia detection using an SE-enhanced DenseNet121 model. The main contributions are as follows:

- We develop an SE-enhanced DenseNet121 [7] model for privacy-preserving pneumonia detection under federated learning.
- We compare the performance of FedAvg and FedProx using the RSNA Pneumonia Detection Challenge dataset [8] partitioned into three simulated hospital clients.
- We evaluate the trained models on an independent chest X-ray dataset [12] to assess cross-domain generalization.
- We employ Grad-CAM [6] to provide visual explanations and improve model interpretability.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework. Section IV describes the experimental setup. Sec-

tion V presents the experimental results, Section VI discusses the findings, and Section VII concludes the paper.

II. RELATED WORK

A. CNNs for Chest X-Ray Classification

DenseNet121 [3] has become a widely adopted architecture for chest X-ray analysis because its dense connectivity promotes feature reuse while maintaining high accuracy with relatively few parameters. CheXNet [2] demonstrated its effectiveness for automated pneumonia detection using the ChestX-ray14 dataset [9], while the RSNA Pneumonia Detection Challenge [8] later introduced a more clinically curated benchmark with expert annotations. However, these studies primarily focus on centralized training and do not address distributed learning across healthcare institutions.

B. Attention Mechanisms

Squeeze-and-Excitation (SE) blocks [7] improve feature representation by adaptively recalibrating channel-wise responses with minimal computational overhead. Their ability to emphasize informative features makes them well suited for chest X-ray analysis, where pathological regions often occupy only a small portion of the image. Despite their success in medical image classification, their integration with federated learning remains relatively unexplored.

C. Federated Learning in Medical Imaging

Federated Learning, introduced by McMahan *et al.* [4], enables collaborative model training without sharing raw patient data. FedAvg provides efficient model aggregation but is sensitive to heterogeneous client distributions. FedProx [5] addresses this limitation by introducing a proximal regularization term that stabilizes local optimization. In medical imaging, Sheller *et al.* [10] demonstrated the feasibility of multi-institutional federated learning, while Kaissis *et al.* [11] highlighted its importance for privacy-preserving healthcare AI. However, direct comparisons between FedAvg and FedProx for pneumonia detection using the same backbone are still limited.

D. Explainability

Explainability is essential for increasing clinical confidence in deep learning models. Grad-CAM [6] is one of the most widely adopted post-hoc visualization techniques because it highlights image regions influencing model predictions without modifying the network architecture. Although extensively used for centralized models, its application to compare different federated optimization strategies has received limited attention.

E. Research Gap

Existing studies typically investigate network architecture, federated optimization, or explainability independently. Few integrate attention-enhanced CNNs, federated learning, and explainable AI within a single framework for pneumonia detection. In addition, cross-dataset evaluation is rarely reported,

TABLE I
ARCHITECTURAL AND SYSTEM CONFIGURATION UNDERLYING FIG. 1

Component	Configuration
Backbone	DenseNet121 (ImageNet-pretrained)
Attention placement	One SE block after each dense block
Classification head	Global average pooling $\rightarrow$ FC $\rightarrow$ Sigmoid
Federated clients	3 simulated hospitals
Local optimizer	Adam ($E = 2$ local epochs/round)
Aggregation	FedAvg or FedProx ($\mu = 0.01$)
Communication	Model weights, 50 rounds
Interpretability	Grad-CAM (last dense block)

despite being essential for assessing model robustness in real-world clinical deployment. This work addresses these gaps through a unified and systematically evaluated framework.

III. METHODOLOGY

Fig. 1 presents the proposed framework, which consists of four stages: (i) centralized training of DenseNet121, (ii) enhancement with Squeeze-and-Excitation (SE) attention, (iii) federated training of the SE-DenseNet121 model across three simulated hospitals using FedAvg or FedProx, and (iv) Grad-CAM-based interpretation of the trained global models.

The overall framework is designed to improve feature representation, preserve patient privacy during collaborative learning, and provide interpretable predictions suitable for clinical decision support.

A. SE-Enhanced DenseNet121

DenseNet121 [3] is selected as the backbone because its dense connectivity encourages feature reuse while maintaining high classification accuracy with relatively few parameters. To improve feature representation, an SE block [7] is inserted after each dense block, allowing the network to adaptively emphasize informative feature channels.

Given an intermediate feature map $\mathbf{U} \in \mathbb{R}^{C \times H \times W}$, the squeeze operation computes channel-wise descriptors as

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j), \quad (1)$$

which are transformed through the excitation module,

$$\mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})), \quad (2)$$

where δ denotes ReLU and σ is the sigmoid activation. The resulting attention weights recalibrate each channel,

$$\tilde{u}_c = s_c u_c, \quad (3)$$

before feature propagation. The network performs binary classification using a sigmoid output layer optimized with class-weighted binary cross-entropy loss.

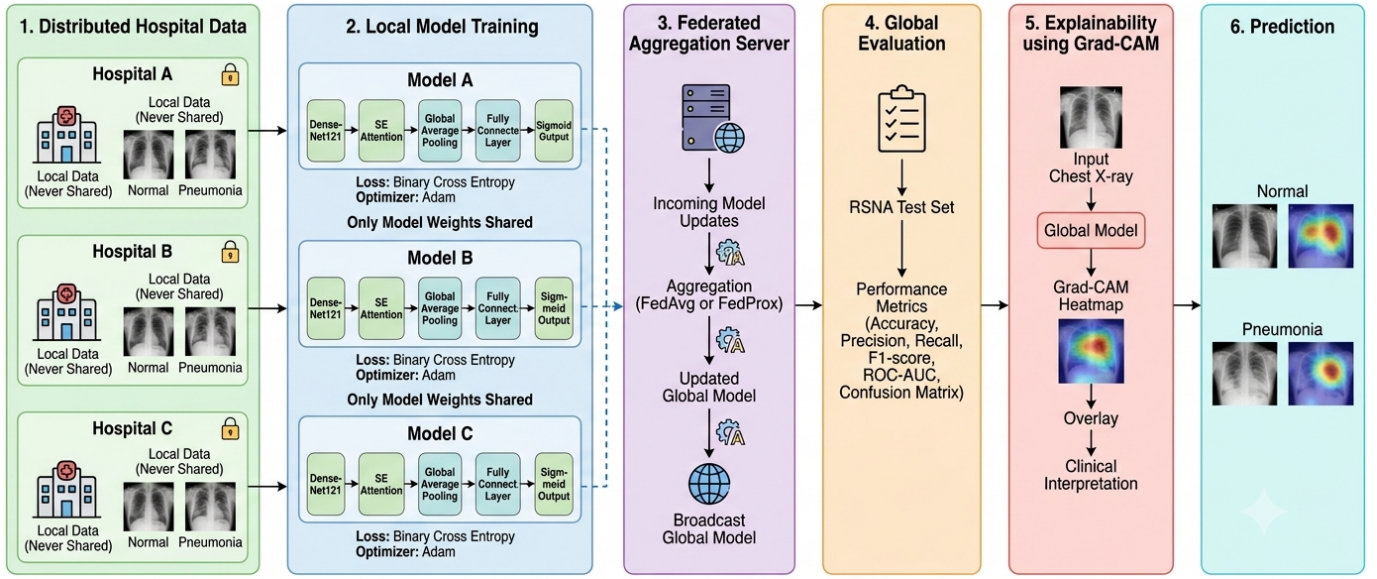


Fig. 1. Overview of the proposed Explainable Federated Learning framework. Local SE-DenseNet121 models are trained across three simulated hospitals and aggregated using FedAvg or FedProx. The global model is evaluated on the RSNA test set and an independent Chest X-ray dataset with Grad-CAM explanations.

B. Federated Learning

The RSNA training set is partitioned into three non-overlapping client datasets representing independent hospitals. During each communication round, the server distributes the global model to all clients, each client performs two local training epochs, and the updated parameters are returned for aggregation.

FedAvg [4] computes the global model using sample-weighted averaging,

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1}, \quad (4)$$

where $n = \sum_k n_k$. FedProx [5] uses the same aggregation strategy but modifies the local objective by introducing a proximal regularization term,

$$\min_w F_k(w) + \frac{\mu}{2} \|w - w^t\|^2, \quad (5)$$

which reduces client drift under heterogeneous data distributions. The overall federated training procedure is summarized in Algorithm 1.

C. Grad-CAM Visualization

Model interpretability is evaluated using Grad-CAM [6]. For a target class c , the importance of feature map A^k is computed as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (6)$$

and the localization map is generated as

Algorithm 1 Federated Training Procedure

- 1: Initialize global model w^0
- 2: **for** $t = 1$ to T **do**
- 3: Broadcast w^{t-1} to all clients
- 4: **for all** clients **do**
- 5: Perform local training for E epochs
- 6: Return updated model parameters
- 7: **end for**
- 8: Aggregate client models using FedAvg or FedProx
- 9: **end for**
- 10: **return** w^T

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right), \quad (7)$$

highlighting image regions that contribute most to the model's prediction. Grad-CAM visualizations are generated for correctly and incorrectly classified images to qualitatively assess the consistency of model attention.

IV. EXPERIMENTAL SETUP

A. Datasets

The RSNA Pneumonia Detection Challenge dataset [8] is used for training, validation, and internal testing using a stratified 70/15/15 split. The Labeled Chest X-Ray Images dataset [12] is reserved exclusively for external evaluation to assess cross-domain generalization (Table II).

B. Preprocessing

All images are resized to 224×224 and normalized using ImageNet statistics. Training images are augmented with ran-

TABLE II
DATASETS USED IN THIS STUDY

Dataset	Purpose	Classes
RSNA Pneumonia Detection Challenge	Train/Validation/Test	2
Labeled Chest X-Ray Images	External Evaluation	2

TABLE III
TRAINING HYPERPARAMETERS

Parameter	Value
Backbone	DenseNet121 + SE
Optimizer	Adam
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Learning rate scheduler	ReduceLROnPlateau
Input size	224×224
Batch size	32
Loss function	Class-weighted Binary Cross-Entropy
Mixed precision	Enabled (AMP)
Random seed	42
Clients	3
Local epochs	2
Communication rounds	50 (Early stopping)
Aggregation	FedAvg, FedProx

TABLE IV
HARDWARE AND SOFTWARE CONFIGURATION

Component	Specification
Platform	Kaggle Notebooks
GPU	NVIDIA Tesla T4 (16 GB)
Programming language	Python 3
Framework	PyTorch
Precision	Automatic Mixed Precision (FP16/FP32)

dom horizontal flips, small rotations, and brightness/contrast adjustments, while no augmentation is applied during evaluation.

C. Training Configuration

All models are implemented in PyTorch using an ImageNet-pretrained DenseNet121 backbone. Training uses the Adam optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . Learning rate scheduling is performed using ReduceLROnPlateau with automatic mixed-precision training. Each federated client performs two local epochs per communication round for a maximum of 50 rounds, with early stopping based on global validation loss. Training hyperparameters and system specifications are summarized in Tables III and IV.

D. Client Partitioning

The RSNA training set is partitioned into three disjoint client datasets representing independent hospitals. During federated training, only model parameters are exchanged with the central server, while patient images and labels remain on their respective clients.

E. Evaluation Metrics

Performance is evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. Confusion matrices, ROC curves, and global validation accuracy and loss are also analyzed throughout federated training. Four models are compared: (1) centralized DenseNet121, (2) centralized DenseNet121+SE, (3) federated DenseNet121+SE using FedAvg, and (4) federated DenseNet121+SE using FedProx.

V. RESULTS

A. Internal Test Performance (RSNA)

Table V summarizes the performance of the proposed federated models on the RSNA held-out test set. FedAvg achieved an accuracy of 82.49% with an AUC of 88.52%, while FedProx obtained a comparable accuracy of 82.16% and a slightly higher AUC of 88.63%. Although both methods produced similar overall performance, FedProx achieved higher recall (77.27% vs. 77.27%) with a small reduction in precision (57.79% vs. 58.96%), indicating fewer missed pneumonia cases at the cost of slightly more false positives. Overall, both models performed consistently, with FedProx providing a more clinically desirable balance.

TABLE V
CENTRALIZED VS. FEDERATED PERFORMANCE ON THE RSNA INTERNAL TEST SET

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
Centralized D121	83.14	61.18	68.85	64.79	88.54
Centralized D121+SE	83.99	63.41	68.4	65.81	88.5
Federated, FedAvg	82.49	58.96	73.28	65.35	88.52
Federated, FedProx	82.16	57.79	77.27	66.13	88.63

B. Cross-Dataset Generalization

Table VI presents the external evaluation on the independent Labeled Chest X-Ray Images dataset. FedAvg maintained very high precision (99.03%) but achieved low recall (26.15%), indicating that many pneumonia cases were missed. In contrast, FedProx substantially improved recall (54.62%) while maintaining comparable precision (96.82%) and a slightly higher AUC (90.35%). Consequently, the F1-score increased from 41.38% to 69.84%, demonstrating better robustness and cross-domain generalization.

TABLE VI
EXTERNAL GENERALIZATION ON THE INDEPENDENT LABELED CHEST X-RAY IMAGES DATASET, NO FINE-TUNING

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
Centralized D121	53.2	99.00	25.38	40.41	79.8
Centralized D121+SE	45.99	98.18	13.85	24.27	88.82
Federated, FedAvg	53.69	99.03	26.15	41.38	88.88
Federated, FedProx	70.51	96.82	54.62	69.84	90.35

C. Convergence

Figs. 2 and 3 illustrate the global validation accuracy and loss during federated training using FedAvg and FedProx, respectively. Both methods show stable convergence, with validation accuracy improving rapidly during the initial communication rounds and the validation loss decreasing consistently. FedAvg converges around communication round 30, where early stopping terminates training, whereas FedProx reaches stable performance by approximately round 20. This demonstrates that FedProx achieves comparable performance with fewer communication rounds, reducing communication overhead.

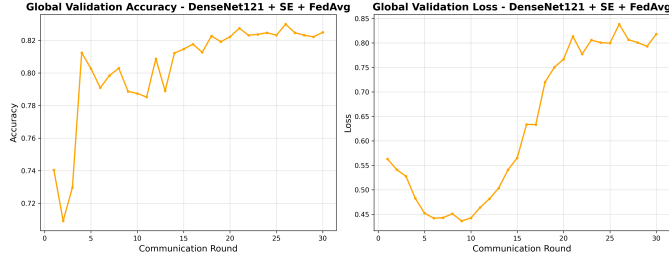


Fig. 2. Global validation accuracy and loss during federated training using FedAvg.

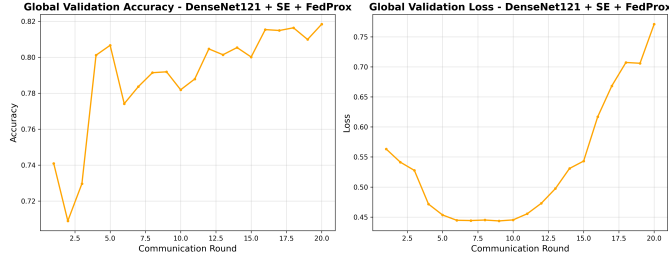


Fig. 3. Global validation accuracy and loss during federated training using FedProx.

D. Confusion Matrices and ROC Curves

Figs. 4 and 5 present the normalized confusion matrices for the FedAvg and FedProx models on the RSNA test set. Both models achieve strong classification performance, with FedProx correctly identifying a larger proportion of pneumonia cases, consistent with its higher recall reported in Table V. The confusion matrices also show that the improved sensitivity of FedProx is accompanied by a small increase in false-positive predictions, reflecting the typical trade-off between recall and precision. The corresponding ROC curves are shown in Fig. 6(a) and Fig. 6(b). Both models achieve AUC values above 88%, indicating reliable discrimination across different decision thresholds and consistent classification performance. Overall, these results demonstrate that both federated models provide robust diagnostic capability, with FedProx offering a slight advantage in pneumonia detection while maintaining competitive overall performance.

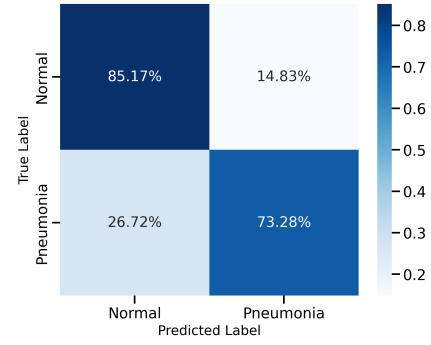


Fig. 4. Normalized confusion matrix of the FedAvg model on the RSNA test set.

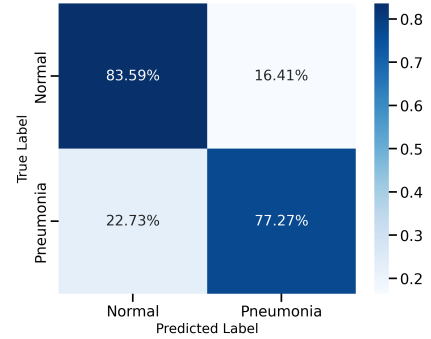
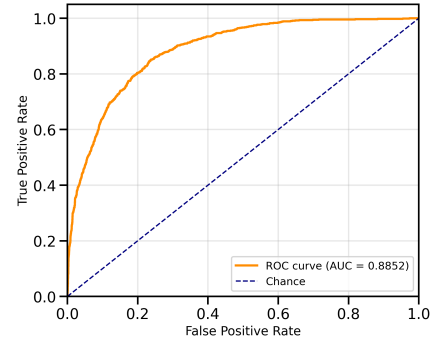
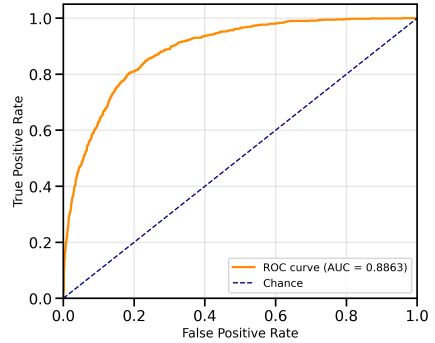


Fig. 5. Normalized confusion matrix of the FedProx model on the RSNA test set.



(a) FedAvg (AUC = 88.52%).



(b) FedProx (AUC = 88.63%).

Fig. 6. ROC curves of the FedAvg and FedProx models on the RSNA test set. Both models achieve comparable discriminative performance, with AUC values exceeding 88%.

E. Grad-CAM Explainability

Fig. 7 compares Grad-CAM visualizations generated by the FedAvg and FedProx global models on representative RSNA test images. For correctly classified normal and pneumonia cases, both models focus on clinically relevant pulmonary regions, indicating meaningful feature learning. In the representative misclassified example, the activation extends beyond the pathological region into surrounding anatomical structures, suggesting the influence of overlapping ribs or mediastinal regions. Overall, both federated models maintain interpretability while providing clinically meaningful explanations for their predictions.

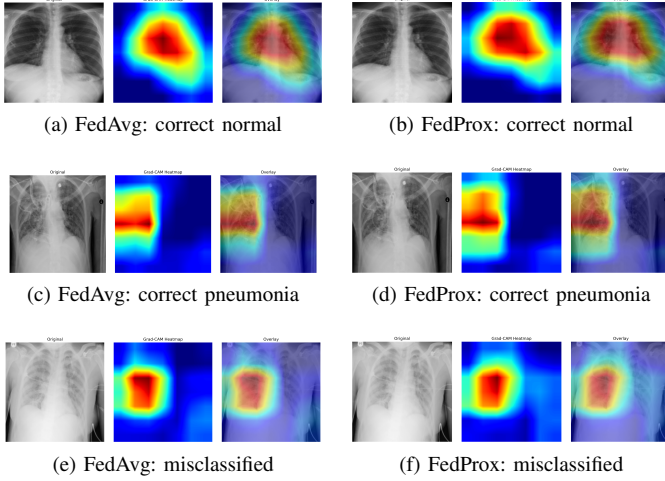


Fig. 7. Grad-CAM comparison, FedAvg vs. FedProx: (a)–(b) correctly classified normal cases, (c)–(d) correctly classified pneumonia cases, (e)–(f) representative misclassified pneumonia cases. Correct predictions concentrate on clinically relevant pulmonary regions, while misclassified examples show broader activation beyond expected lesion boundaries.

VI. DISCUSSION

Although FedAvg and FedProx achieved similar performance on the RSNA test set, their behavior differed considerably on the external dataset. FedProx maintained substantially higher recall (54.62% vs. 26.15%), indicating improved robustness and cross-domain generalization to unseen data, which is important for real-world clinical deployment where imaging conditions often differ from the training distribution. This improvement is likely due to the proximal regularization in FedProx, which constrains local model updates and reduces client-specific overfitting during federated optimization [5]. Grad-CAM [6] visualizations further showed that correctly classified images focused on clinically relevant lung regions, whereas misclassified cases exhibited attention outside the pathological areas, improving model interpretability despite providing only qualitative explanations. Nevertheless, this study has several limitations, including the use of simulated clients from a single dataset, binary pneumonia classification, and the absence of complete centralized baseline comparisons. Future work will validate the proposed framework on indepen-

dent multi-institutional datasets to further assess its robustness and generalizability.

VII. CONCLUSION

This paper presented an explainable federated learning framework for privacy-preserving pneumonia detection from chest radiographs using an SE-enhanced DenseNet121 model. The framework combines federated optimization with Grad-CAM to provide accurate and interpretable predictions while preserving data privacy. Experimental results showed that FedAvg and FedProx achieved similar performance on the RSNA test set, whereas FedProx demonstrated better generalization on an independent external dataset, particularly in recall, highlighting its potential for real-world clinical deployment. Grad-CAM visualizations further confirmed attention to clinically relevant lung regions, improving prediction transparency. Although the federated clients were simulated from a single dataset, the results demonstrate the potential of explainable federated learning for collaborative medical AI. Future work will evaluate the framework using multi-institutional datasets, extend it to multi-label thoracic disease classification, and incorporate advanced privacy-preserving techniques such as differential privacy and secure aggregation.

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